

Kevin Zhang

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Research interests

Optimal transport, stochastic differential equations, machine learning, artificial intelligence, scRNA-seq, generative modelling, anomaly detection, game theory

Education

- 2019 – 2024 **University of Toronto** – Toronto, Ontario, Canada
PhD in Statistics
Advisors: [Dr. Dehan Kong](#), [Dr. Zhaolei Zhang](#).
- 2015 – 2019 **University of Toronto** – Toronto, Ontario, Canada
HBSc in pathobiology, statistics and mathematics. *major GPA: 4.0/4.0. GPA: 3.9/4.0.*

Publications

- 2023 **Modeling Single Cell Trajectory Using Forward-Backward Stochastic Differential Equations**
[K. Zhang](#), J. Zhu, D. Kong, and Z. Zhang.
PLOS Computational Biology (in revision). [\[Link to preprint\]](#)[\[Link to GitHub\]](#)
- 2023 **Modeling Cell Type Development Trajectory using Multinomial Unbalanced Optimal Transport**
[K. Zhang](#), J. Zhu, Z. Zhang, and D. Kong.
Journal of the American Statistical Association (under review). [\[Link to GitHub\]](#)
- 2023 **LLOT: application of Laplacian Linear Optimal Transport in spatial transcriptome reconstruction**
J. Zhu, [K. Zhang](#), Z. Zhang, and D. Kong.
Journal of the American Statistical Association (under review).

2018 **Conserved *pbp1*/ataxin-2 regulates retrotransposon activity and connects polyglutamine expansion-driven protein aggregation to lifespan-controlling *rdna* repeats.**

L. Ostrowski, A. Hall, K. Szafranski, R. Oshidari, K. Abraham, J. Chan, C. Krustev, K. Zhang, A. Wang, Y. Liu, R. Guo, and K. Mekhail.

Nature: Communications Biology.

Teaching experience

Fall **University of Toronto (Teaching Assistant)** – Toronto, Ontario, Canada
2019-Present Courses include:
STA130: An Introduction to Statistical Reasoning and Data Science
STA220: The Practice of Statistics I
STA255: Statistical Theory
STA261: Probability and Statistics II
STA302: Methods of Data Analysis I
STA303: Methods of Data Analysis II
STA304: Surveys, Sampling and Observational Data
STA305: Design and Analysis of Experiments
STA347: Probability
STA437: Methods for Multivariate Data
STA414: Statistical Methods for Machine Learning II
STA442: Methods of Applied Statistics

Review experience

2023 Canada Communicable Disease Report (2 times)
2023 Journal of the American Statistical Association (1 time)
2022 Journal of the Royal Statistical Society: Series C (1 time)

Presentations

December 2022 Modelling Cellular Development Trajectory using Unbalanced Optimal Transport
Medicine by Design's 7th Annual Symposium
July 2022 Modelling Single-Cell Type Trajectory Using Optimal Transport
The Fifth ICSA-Canada Chapter Symposium

Honors and scholarships

2015-2019	Dean's List (University of Toronto)
2018	New College Council In-Course Scholarship (University of Toronto)
2017	LMP Undergraduate Summer Research Program UROP Award (University of Toronto)
2015	Undergraduate Entrance Award (University of Toronto)

Technical skills

Programming languages

Proficient in: Python, R, SQL

Familiar with: C++, Linux/UNIX environment, MATLAB, HTML

Libraries/Packages

Python: NumPy, Pandas, matplotlib, PyTorch (including CUDA GPU Configuration), TensorFlow, scikit-learn, XGBoost

R: dplyr, ggplot2, tidyverse

Software

Git, SLURM Workload Manager, Docker

References

Dr. Dehan Kong

University of Toronto

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Dr. Zhaolei Zhang

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Dr. Piotr Zwiernik

University of Toronto

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